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Petrophysics of Tight Reservoirs - Challenges and Opportunities

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Abstract

Tight reservoirs pose a lot of challenges in formation evaluation due to low porosity and permeability, complex fluid fill driven by drainage and imbibition cycles, variation in rock types, long transition zones and uncertainty in determining free water level. Due to the uncertainty in reservoir properties and fluid fill, it is challenging to select completion zones to maximize hydrocarbon production. Integration of core, open hole & cased hole logs and well test is required to narrow down the uncertainty in reservoir properties and to predict zones which are prone to water production and/or do not contribute to the fluid flow. Water zones are identified by drainage and imbibition log response calibrated with residual hydrocarbon measured on core plugs. Reservoirs with bitumen pose a challenge to accurately quantify its presence and impact on reservoir producibility. Due to the presence of bitumen in the reservoir, which can be in the form of pore lining, pore fill and/or pore throat blocking, reservoir connected porosity and permeability is impaired. If volume of bitumen is not correctly quantified then hydrocarbon in place, completion decisions, reservoir producibility and ultimate recovery is impacted. Basic logs often do not have sufficient contrast to differentiate bitumen from hydrocarbon in the reservoir, hence, advanced logs viz. NMR and spectroscopy logs are required to identify bitumen and its quantification. This paper addresses challenges seen in tight reservoirs through integration of core and logs to understand cause of rock type variation, determining accurate water saturation with improved saturation exponent measured in tight rocks and calibration of log and core NMR, identify water producing zones through integration of open hole and cased hole logs, prediction of low permeability zones with sonic log-based workflows, and identify bitumen rich zones to assess the impact on petrophysical properties.

Introduction

This paper describes various issues pertinent to the reservoirs with low porosity (avg 8-10%) and permeability (<1 mD) often classified as tight reservoirs. These reservoirs often have challenges related to accurate quantification of porosity and water saturation from the static properties point of view, and permeability and producibility from dynamic point of view. Additional challenges like free water level estimation, issues with bitumen in the pore space, early water production, and frac height detection are few examples which often complicate the evaluation of these tight reservoirs. In this paper, we try to address some of these challenges through proper data acquisition and/or including the advanced logs information to

draw more insights which could help to reduce the underlying uncertainties in the evaluation. Concentrated efforts should be done to maximize open hole and cased hole log data acquisition and conventional core in the first few wells to understand the cause of reservoir properties variation, to calibrate the petrophysical/geomechanical model, and producibility/fluid phase of each perforated/fracked reservoir zone.

Bitumen (Link to Permeability)

Bitumen presence in pore space of reservoir rocks pose a challenge for hydrocarbon in place calculation and producibility of the rocks decreased significantly depending on the amount of bitumen. The main challenge is to quantify the bitumen in the pore space - this can be solved by acquiring advanced logs viz. Nuclear Magnetic Resonance and Spectroscopy logs along with the core. An integrated model with logs and core can be created to quantify bitumen volume in the pore space.

As bitumen volume varies vertically and laterally, the main challenge is to propagate bitumen volume in-between the wells. Due to the random/patchy nature of bitumen distribution in the reservoir rocks, a kriging approach does not work very well in this case. A soft constraint of bitumen variation between the wells should be derived either from the geological understanding of the bitumen variation in the area and/or seismic attributes/quantitative interpretation should be used to integrate the understanding. A rock physics feasibility study should be done at the well locations to decipher if post stack seismic attribute (acoustic impedance) or pre-stack seismic attributes (Poisson's ratio, lambda-rho, mu-rho) are suitable to predict the volume of bitumen in the reservoir rocks.

As bitumen poses a challenge for reservoir producibility, the volume of bitumen in the pore space should be linked to the permeability of the rock. A detailed workflow to assess the impact of bitumen on permeability has been outlined in (Singhal 2024).

Figure 1 below shows the logs panel alongwith thin section and MICP at few depth points. Bottom sand is showing a severe plugging of pore space by bitumen resulting in a much higher capillary entry pressure when compared to the thin sections and capillary curves for the sand at xx75 m. Due to the presence of bitumen, the producibility of the bottom sand is impacted the most, while the shallower sands indicate production of restricted gas flow along with frac water.

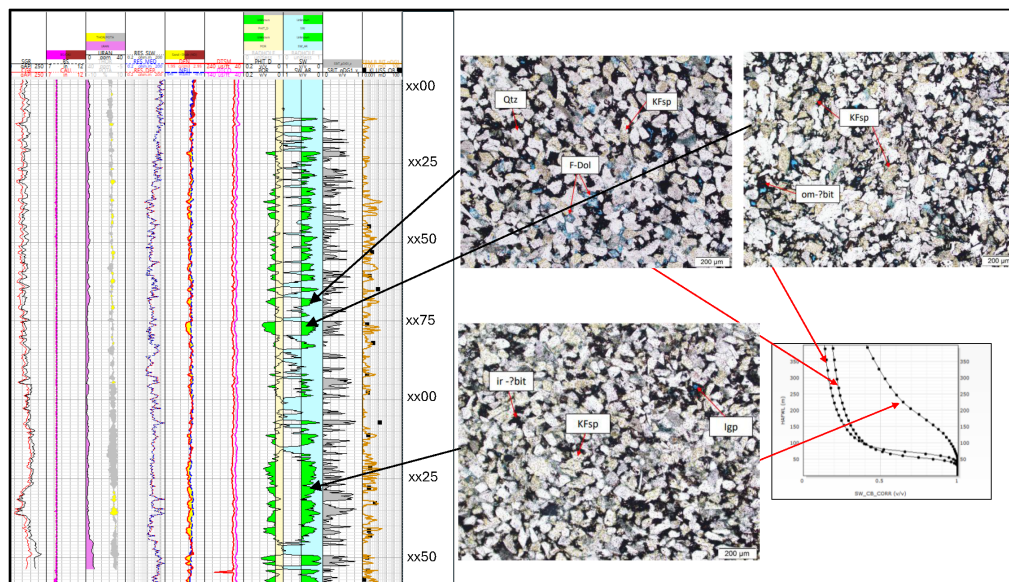


Figure 1—Layout: Track -1: Combined GR (Red) and SGR, Track-2: BS and CALI, Track-3: K, TH and U, Track-4: Shallow (black), Medium (blue) and Deep Resistivity, Track-5: Neutron-Density, Track-6: Compressional and Shear Slowness, Track-7: Porosity, Track- 8: Water Saturation, Track-9: Bitumen Saturation, Track-10: Log and Core permeability. Right: Thin Sections showing the presence of bitumen in the pore space.

Identification of fluid type (Drainage versus Imbibition)

Due to low porosity and permeability, rocks have high capillary entry pressure which causes these rocks to have high water saturation even at high above free water level. If better porosity and permeability rocks are intercalated between lower porosity and permeability rocks, then a complex fluid fill scenario may result. Fluid fill scenario can even be more complicated if imbibition has been observed in the reservoir rocks. In geological times, there are several causes that could cause imbibition of water into the gas bearing reservoir viz. tilting of the structure, hydrodynamic aquifer etc. are some of the causes. As water imbibes into the rock, a residual hydrocarbon is left behind in the pore space which is often observed on resistivity logs. Even in the low porosity and permeability rocks, resistivity-based water saturation does indicate presence of residual hydrocarbon, and one can identify the present-day fluid contact based on saturation profile. To quantify the water saturation accurately from resistivity, saturation exponent is one of the key parameters that should be underpinned for the range of water saturation seen in the reservoir rocks. Even though, there may be the same porosity of reservoir rock across the present-day fluid contact, there may be a variation of saturation exponent which depend on water saturation itself. SCAL PCRI experiments are required to determine the variation of saturation exponent with saturation, and with different rock types.

Once water saturation and present-day fluid contacts have been established in the reservoir, the next challenge is to propagate water saturation in the reservoir model with a suitable saturation height function. As imbibition has been observed in the reservoir rocks, a drainage capillary pressure-based saturation height function may not capture the water saturation as calculated by resistivity profile. If there are no imbibition capillary pressure measurements are available, then saturation height function can be derived using resistivity derived water saturation, porosity (and/or permeability) of the rock and present-day fluid contact. Sometimes, the shape of the log derived saturation height function does not match with actual reservoir rock capillary pressure curves (imbibition) resulting in an early/late water breakthrough while doing reservoir production history match in dynamic model. Shape of the log derived saturation height function should be modified iteratively to make sure transition zone behavior is not too far from the actual core measured capillary curves while honoring the log water saturation and production behavior.

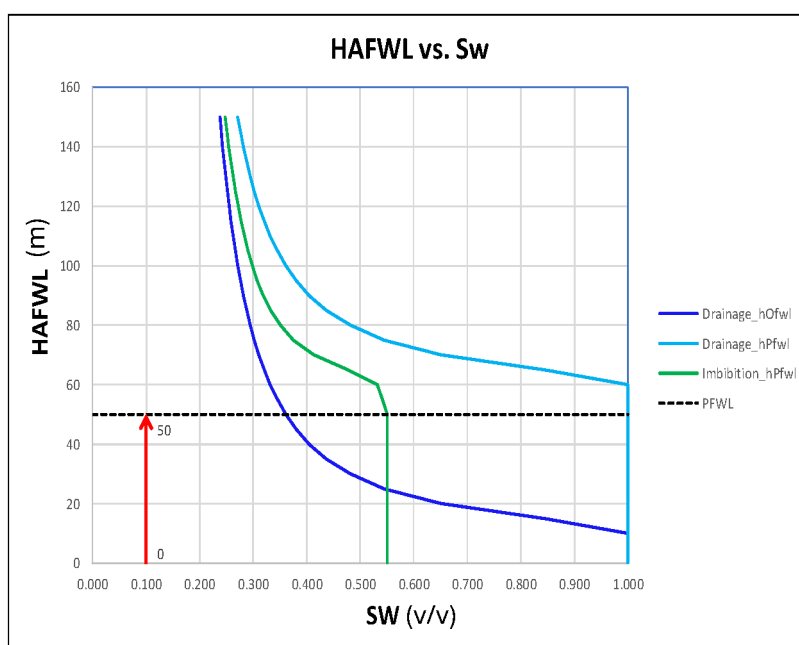


Figure 2—Capillary pressure converted to Height Above Free Water Level (HAFWL) as shown on y-axis. Drainage process is indicated by blue curve. Imbibition is represented by green curve, while cyan curve represents the drainage curve when applied to present day contact which is 50 m above the initial FWL as shown.

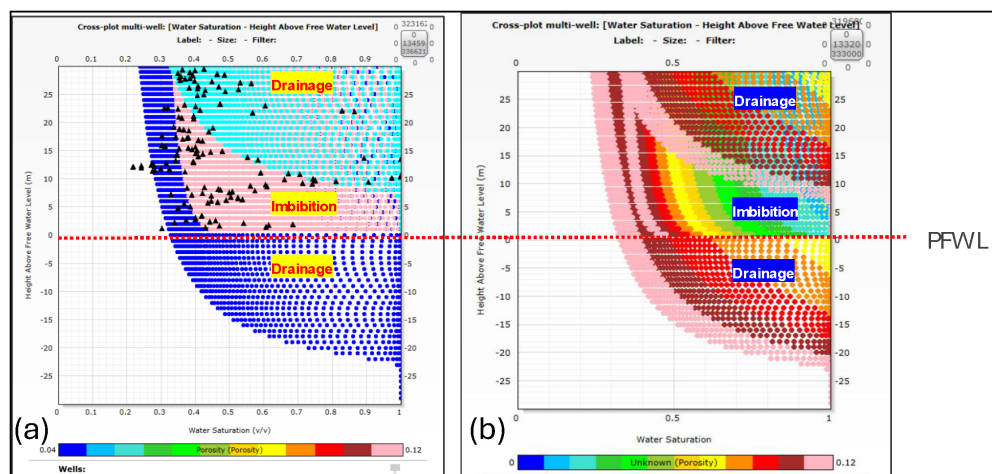


Figure 3—Drainage and Imbibition models: (a) Drainage and Imbibition models are shown along with actual log data points. Imbibition model (pink) captures most of the actual log data points; (b) Drainage and Imbibition models color coded with porosity classes.

Detection of water/HC production through PLT sensors

When low porosity and permeability rocks are intercalated with good quality rocks, hydraulic fracturing of such rocks pose a challenge for water production. Generally, with more pressure drawdown water rate increases and at times with basic PLT sensors it is difficult to know where the high percentage of water is coming from when more than one zones are fracked and flowed commingled. Fluid density and capacitance sensors sometimes do not detect small amount of water produced in high gas rate wells. In the high-water rate wells, capacitance sensor gets saturated above 40% water holdup. Advanced PLT sensors are required to fully characterize the fluid phase in such wells - resistivity, optical density and/or oxygen activation-based sensors are worth considering depending on flow rates and phase of the fluid.

As shown in Figure 4, resistivity image is insensitive to differentiate between oil and gas production from the bottom perforations, gas production can be detected using optical density due to lower refractive index of the gas than liquids.

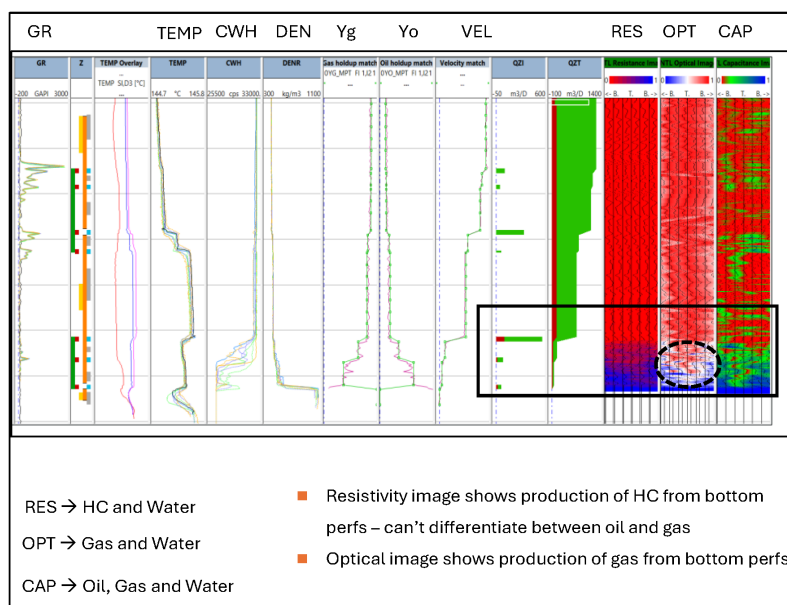


Figure 4—PLT flowing pass across the perforations. Resistivity image (RES) shows production of hydrocarbon from bottom perforations (highlighted by rectangular box). Optical image (OPT) differentiates oil from gas (highlighted by circle).

Fluid Contact Determination

Due to low porosity and permeability of the rock, there has been a little success of acquiring credible formation pressure in such rocks. Most of the time, rocks do not respond to the drawdown provided by downhole formation tester resulting no success to the acquisition of formation pressure. Even if there is a response from the formation, formation pressures tend to be supercharged due to low mobility of the formation. In the absence of formation pressures, it is quite challenging to determine free water level in the tight reservoirs. Fluid fill can be interpreted from logs by looking at change in saturation profile to pick up transition from hydrocarbon to water. However, as explained in the section above, this could be the present-day fluid contact due to imbibition of water. The level of imbibition of water could be different in the different parts of the field (tilting, rock types etc.) which could provide varying fluid contact (if hydrocarbon to water transition is seen at the well locations). In the situation where hydrocarbon to water transition is not seen at the well location, the deepest hydrocarbon down to can be used at the well location to define low case fluid contact. Saturation height function can also be used to iterate free water level such that error between saturation height function-based water saturation and resistivity-based water saturation is minimized. Structure spill point, information of nearby field fluid contact /column heights, seismic quantitative information may also be used to bracket the fluid contact uncertainty.

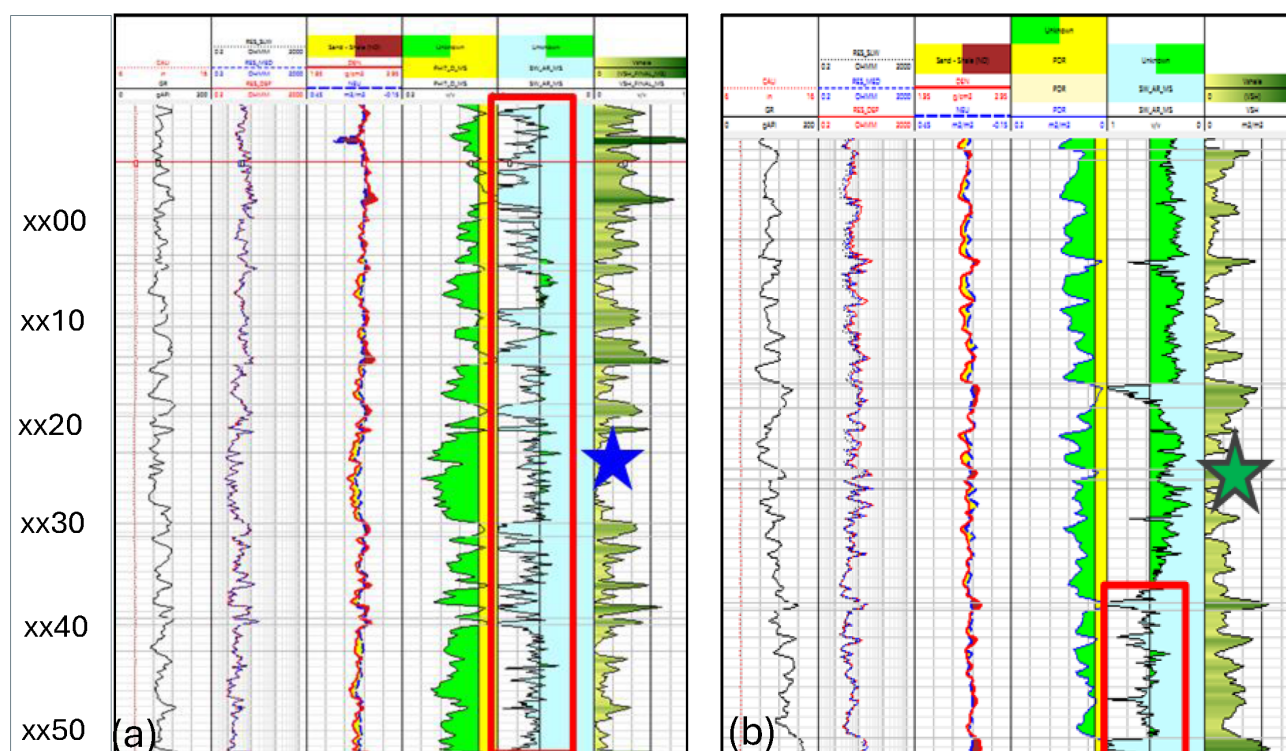


Figure 5—(a) plot showing water saturation is around 0.50-0.55 v/v, blue star indicates downhole formation tester water sample; (b) plot showing sharp cut of the water saturation from 0.70 v/v to 0.50 v/v indicating presence of imbibition in the reservoir sand, green star indicates downhole formation tester gas sample.

Nuclear Magnetic Resonance Derived Rock Types

Nuclear Magnetic Resonance (NMR) T2 distribution captures pore size distribution of the rock. For heterogenous rocks with wide variety of pore sizes, NMR T2 distribution shows broad spectrum with few amplitude peaks at low, mid and high T2 times. However, if there is gas present in the reservoir, then the peak of the NMR T2 amplitude, corresponding to free fluid region, shifts towards left compared to 100% water saturated rock. To fully characterize the pore size distribution and derive facies from the NMR T2 distribution, NMR T2 distribution should be fluid substituted to make it 100% water saturated NMR T2

Figure 10 consists of two panels, (a) and (b), comparing the results of the proposed method with the results of the conventional method.

Panel (a) shows four columns of plots labeled GAUDIS, GAUDIS1, GAUDIS2, and GAUDIS3. Each column contains multiple stacked plots showing the results of the proposed method (colored lines) and the results of the conventional method (black lines). The plots show the results of the proposed method (colored lines) and the results of the conventional method (black lines). The plots show the results of the proposed method (colored lines) and the results of the conventional method (black lines).

Panel (b) shows a comparison of the results of the proposed method with the results of the conventional method. The plots show the results of the proposed method (colored lines) and the results of the conventional method (black lines). The plots show the results of the proposed method (colored lines) and the results of the conventional method (black lines).

Another challenge is to use consistent NMR tool, acquisition sequence, and processing parameters in the wells. Variation of any of these may cause the variation of T2 distribution and hence the variation of rock types in the reservoir which are just the artefacts of the acquisition and processing. A thorough QA/QC should be done for raw NMR echoes and subsequently inversion of these echoes to T2 distribution before NMR T2 distribution can be used for any rock typing or porosity/saturation estimates.

Bulk volume of water in a rock is the total volume of water in the pore space. It includes clay bound water, capillary bound water and free water. In tight and heterogenous rocks, it is often challenging to identify the zones with free water due to the presence of different rock types and uncertainty in free water level, bulk

volume of water approach can be used to identify the zones with free water. Generally, when porosity and saturation are plotted in a cross-plot, a clear trend of decreasing water saturation with increasing porosity has been observed. Rocks with same porosity can have different water saturation depending on permeability of the rock and its position from free water level. At the same height above free water level, low permeability rocks tend to contain more bound water when compared to higher permeability rocks due to higher entry pressure for hydrocarbon to move into the low permeability rock. A careful calibration of bulk water volume with production is required to identify the rocks which produce water. As shown in the saturation-porosity crossplot, bulk volume of water space has been classified with various segments showing rock types with more probability of water production.

Figure 7 shows the rock types generated using porosity and saturation cross-plot. Green, pink and blue stripes are the corridors of rocks having high probability of gas, transition zone, and water respectively with the variation of porosity. Further subdivisions of these stripes are done to define the fluid distribution according to porosity and saturation. Higher numbers are assigned to the rocks with good porosity followed by lower numbers for low porosity rocks. A rock type number has been assigned to each depth based on this subdivision.

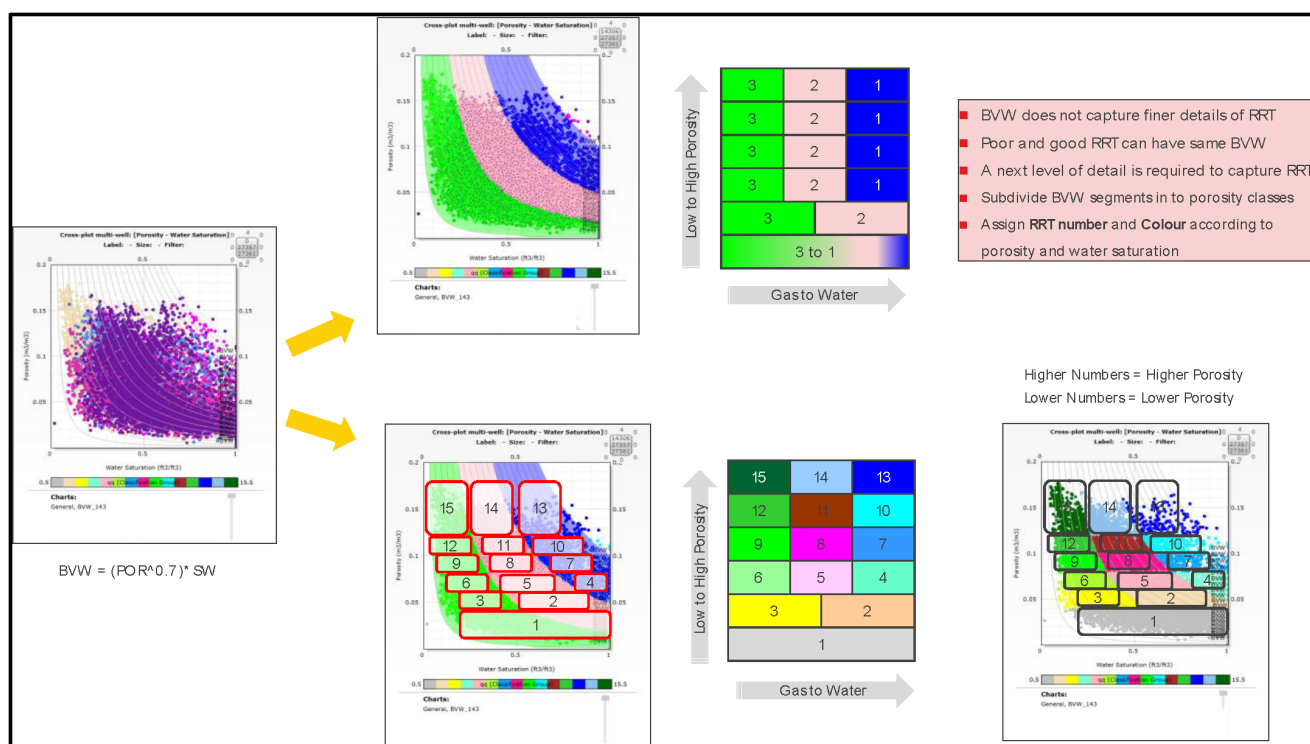


Figure 7—Workflow showing rock types based on porosity and saturation.

Figure 8 shows better porosity rocks are represented by higher numbers. Dark green to dark blue color transition indicate higher to lower gas saturation for good porosity rock. Green color shades indicate transition from high to low porosity in high gas saturation rocks. Blue color shades indicate transition from hi to low porosity in high water saturation rock. Rocks with intermediate gas saturation are indicated by brown and pink colors.

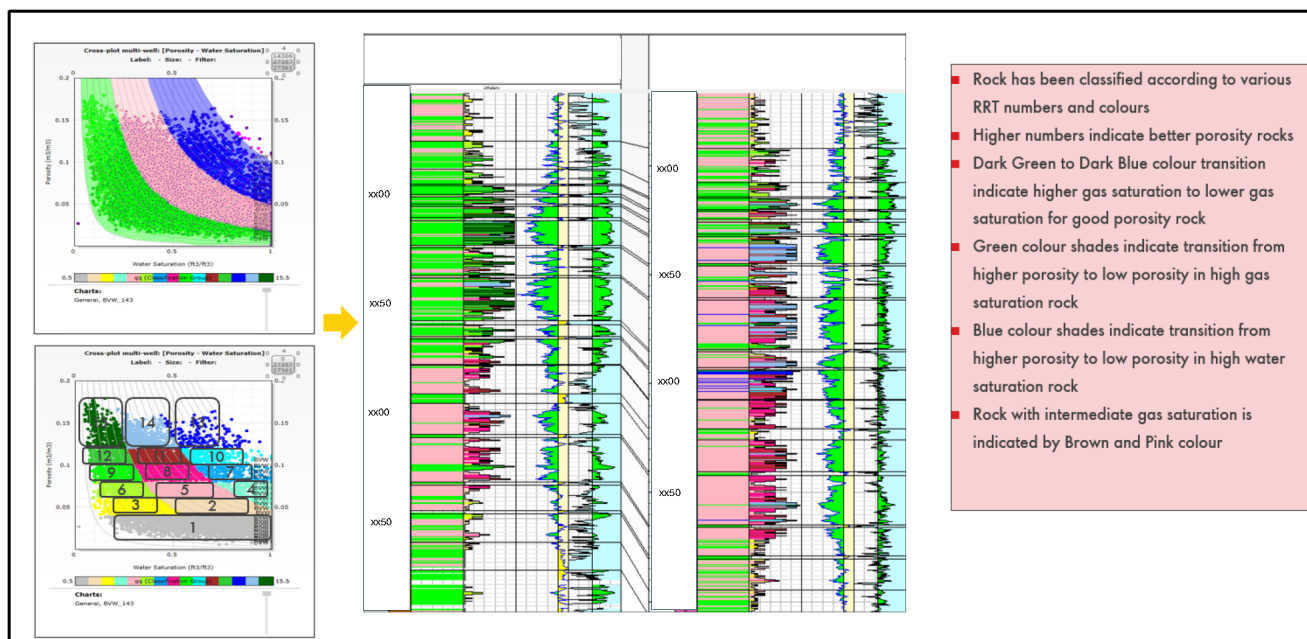
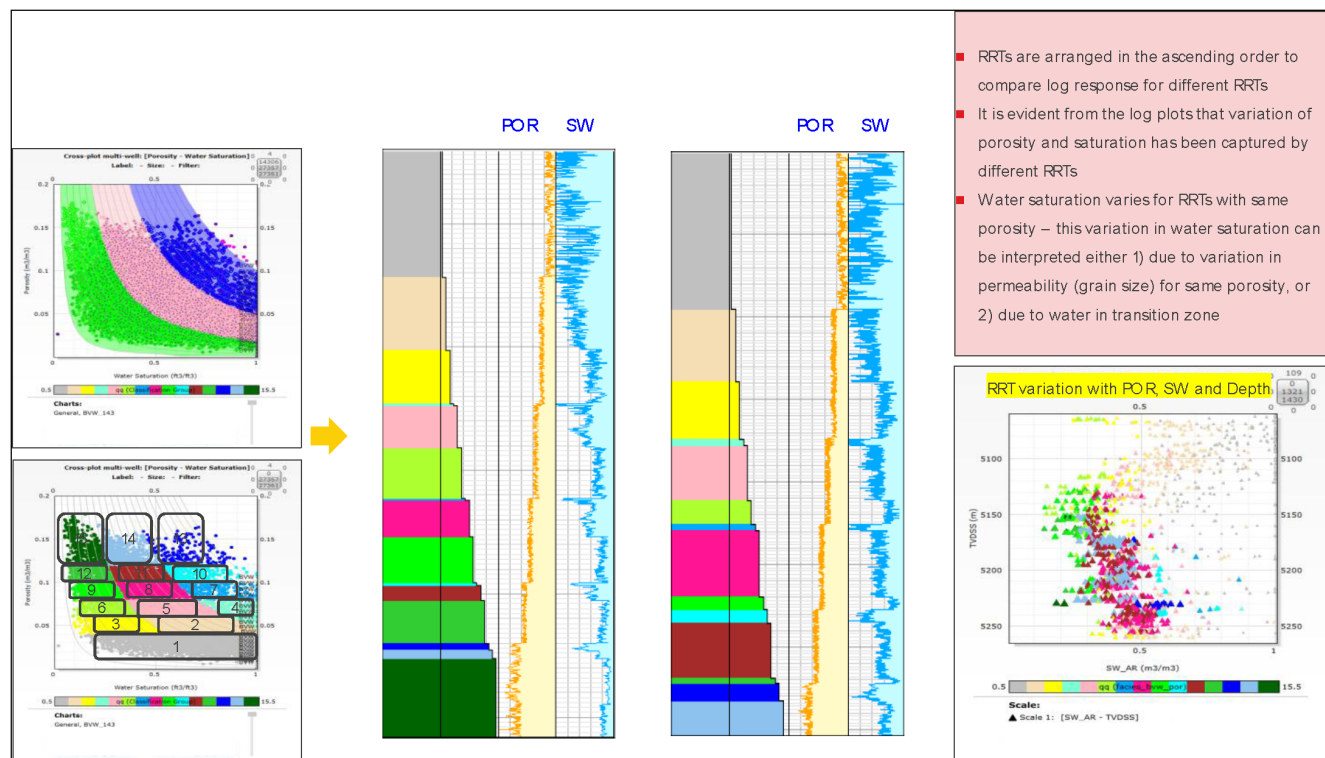


Figure 9 shows the sorted rock types from poor to good as indicated by porosity and saturation log plot. Actual log data has been plotted in TVDSS vs SW crossplot color coded by rock types with size of the point reflect the porosity. This plot reveals the depth variation of various rock types present in the field.



Stoneley Permeability workflow

Matrix permeability is one of the major impacting parameters in tight reservoirs which controls producibility even after fracking these reservoirs. Core permeability can be cross plotted with core porosity to derive a transform which can be applied on log porosity to compute permeability at well locations. However, if there are rock type variations across the field i.e. permeability varies independent of porosity then core porosity-permeability transform will not capture the variation of permeability. NMR is generally used to compute permeability by Coates-Timur and/or SDR method which require calibration with core. If NMR data has low signal to noise ratio or affected due to borehole conditions and/or core calibration is not available, then the permeability derived from NMR will have large uncertainty.

Stoneley waves are generated at liquid-solid boundary in the borehole when monopole transmitter is fired in sonic tool. These waves attenuate against high permeability formation and this information can be used to derive permeability profile along the borehole. Measured Stoneley slowness is compared with modelled Stoneley slowness (computed as a function of mud velocity, formation shear slowness and bulk density) to calculate difference in slowness time which is a qualitative indicator of permeability (Buffin 1996, Kumar 2008). After core calibration this qualitative indicator can be used as permeability of the rock.

Figure 10 shows the stoneley slowness picked from the semblance plot. DTST_DIFF reflects the difference between measured and modelled stoneley slowness. Despite having the same porosity in 2 wells, as shown below, DTST_DIFF has very different values. When integrated with the gas production rate from these 2 wells, lower DTST_DIFF values correspond to lower gas production rate and vice versa.

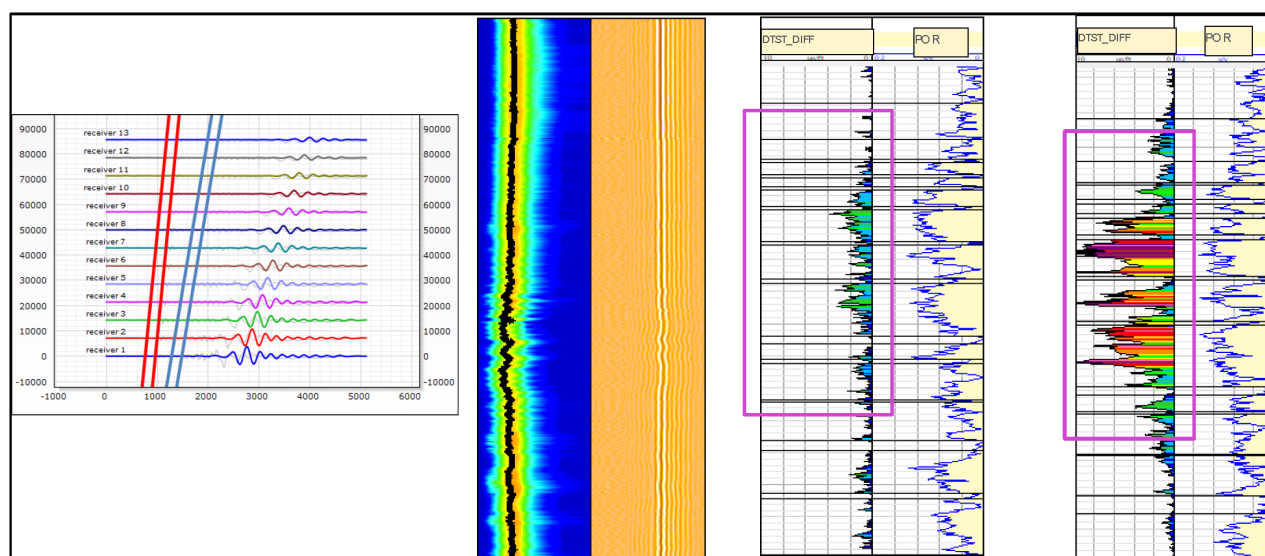


Figure 10—Layout showing the semblance plot for stoneley slowness picking. Right: DTST_DIFF reflect the difference between measured and modelled stoneley slowness.

Digital Rock Analysis - saturation exponent and rel perm estimate

Due to tight nature of the rock, routine and special core analysis takes a lot of time. Digital rock analysis provides an alternative to create digital image of the rock and internal structure which can then be used to simulate reservoir properties (Salmani 2025, Apple 2023). Analysis time can be significantly reduced when compared to actual physical measurement on the core plugs. However, to make sure digital rock analysis is simulating the correct output of reservoir properties, physical measurements are required on few core plugs to verify the digital rock analysis. Saturation exponent is one of the key uncertainties in tight reservoir evaluation especially when reservoir is under imbibition. Log of resistivity index and log of water saturation are plotted to derive saturation exponent as a slope. Generally, clean rocks with insignificant clay proportion have straight line on the plot. However, it has been observed that log of resistivity index and

log of water saturation is a concave down curve most of the time. This may be caused by the presence of microporosity in the tight rocks, presence of clay etc. Hence, saturation exponent (or slope of the curve) changes with the water saturation. When reservoir is under imbibition, due to residual hydrocarbon left in the pore space and/or wettability change, saturation exponent in the imbibition is often different than in the drainage. These drainage/imbibition effects and its impact on saturation exponent can be simulated with digital rock analysis (Apple 2023).

Figure 11(a) shows the RI vs SW crossplot indicating effect of microporosity at lower SW values. The plot is color coded by bitumen saturation for each core sample. Despite having high bitumen saturation in one of the core samples (pink), resistivity index value drops significantly at lower water saturation values. Figure 11(b) shows the variation of saturation exponent with water saturation. Each histogram is filtered by the water saturation value and the corresponding mean saturation exponent is shown. As shown, saturation exponent increases with the increase in water saturation. Petrophysical evaluation with variable saturation exponent can be done to account for its variation.

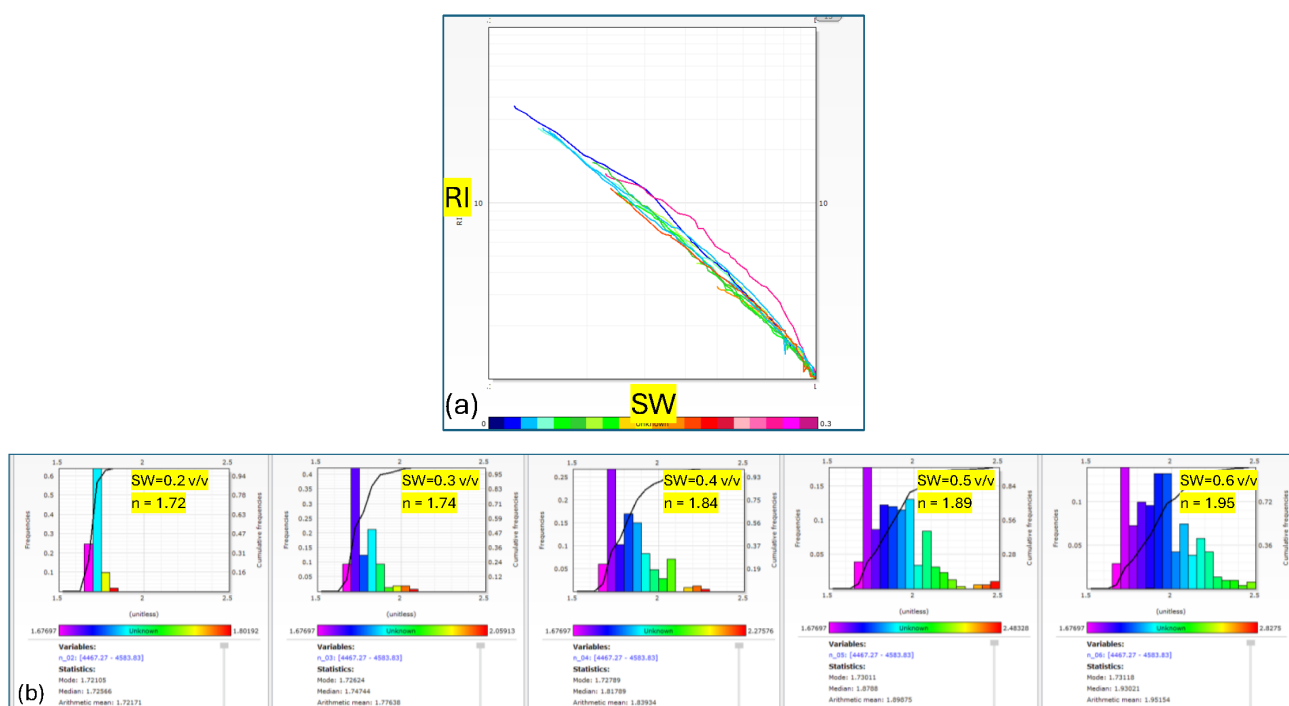


Figure 11—Variation of resistivity index / saturation exponent with water saturation: (a) crossplot between resistivity index vs water saturation (color coded by bitumen saturation) showing effect of microporosity pronounced at lower water saturation; (b) histogram of saturation exponent (color coded by saturation exponent itself) showing the variation of saturation exponent with water saturation.

Conclusion

Tight reservoirs present a multifaceted challenge in petrophysical evaluation due to their inherently low porosity and permeability, complex fluid fill behavior, and the presence of bitumen and other heterogeneities. This paper demonstrates that a robust, integrated approach—leveraging core data, advanced logging techniques, and digital rock analysis—can significantly reduce uncertainty in evaluating these reservoirs. By calibrating saturation models, identifying fluid types through drainage and imbibition responses, and applying advanced workflows such as Stoneley permeability and NMR-based rock typing, operators can make more informed decisions regarding reservoir completion and production strategies. Furthermore, the quantification of bitumen and its impact on permeability, along with the use of bulk volume water and saturation exponent analysis, provides deeper insights into reservoir quality and producibility.

Ultimately, the integration of multidisciplinary data and advanced petrophysical techniques is essential to unlock the full potential of tight reservoirs and optimize hydrocarbon recovery.

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